ProteinVision

Summary Sheet

Molecular Clustering

Quick reference for scaffold-based grouping and Butina clustering.

1. Scaffold based grouping

- Extract a Bemis–Murcko scaffold: keep rings and the atoms and bonds connecting them; remove side chains.
- Preserve atom types, bond types and connectivity. Use a consistent rule for stereochemistry.
- Give each scaffold a canonical key (a consistent representation), then group molecules with identical keys. No similarity cutoff is needed.

Remember: the same scaffold means the same core. Ring-free molecules have no ring scaffold; grouping their empty scaffolds together does not mean chemical similarity.

2 Fingerprints and similarity

- Binary fingerprints encode structural features as 1/0 bits. Rings and side chains can both contribute. Fix the type, length and radius (if applicable).
- Tanimoto similarity T measures how similar two fingerprints are by comparing the bits that are set to 1. $T = c/(a + b - c)$; a = Number of 1-bits in the first fingerprint, b = Number of 1-bits in the second fingerprint, c = Number of positions where both fingerprints have a 1. Shared zeros do not count. T ranges from 0 (none shared) to 1 (identical fingerprints, not necessarily identical molecules).
- Choose a similarity cutoff t: neighbors have $T > t$. With distance $d = 1 - T$, use $d \leq 1 - t$.

3 Butina construction

- Compute all pairwise similarities and identify each molecule's neighbors using the cutoff.
- Count neighbors and rank molecules from most to fewest.
- Select the highest-ranked unassigned molecule as the center. Group it with all its still-unassigned neighbors.
- Mark the cluster assigned and repeat until every molecule is assigned once. A center with no available neighbors forms a singleton.

Ordering: original Butina keeps the initial ranking by neighbor counts. An optional update counts only unassigned neighbors and re-ranks after each cluster.

4 Understanding Butina clusters

- The center is an actual molecule. The cutoff holds between each member and its center, not necessarily between every pair of members.
- The first center to claim a molecule keeps it; there is no later reassignment to a closer center. Selection order can affect results; similar molecules can lie in different clusters.
- A higher similarity cutoff (lower distance cutoff) generally gives smaller, more numerous clusters. There is no universally best cutoff.

5 Using groups in dataset splits

Keep each complete group in just one of training, validation or test. Aim for proportions by molecule count; whole groups may prevent exact splitting percentage targets.

Prepared by the ProVis ProteinVision GmbH (www.protein.vision). Last modified on 25.10.2026.